Case Study - 1

BrightMart is a medium sized online shopping company. Customer uses its website & mobile application to purchase products.

The Company Stores:-

1. Customer Names
2. Email Addresses
3. Phone Numbers
4. Delivery Addresses
5. Passwords
6. Order Information
7. Payment related information

BrightMart has around 100 employees. The company recently hired several employees. To make things easier, the IT department gave employees access to different internal systems.

However, the company has the following security weaknesses:

- Some employees use simple passwords like Bright123.
- Many employees use the same password for multiple systems.
- Software updates are sometimes delayed.
- The company does not regularly conduct employee cybersecurity Training.
- There is no Multi-factor Authentication for some important accounts.
- Employees can access sensitive information that they do not necessarily need for their jobs.

The Incident:
One Monday morning, an employee named Rahul recieves an e-mail.
Subject: Urgent - Your account will be suspended.
The e-mail appears to come in the company's IT department.
It Says:
"Your company account will be suspended today. Click the link below & verify your password immediately."
Rahul clicks the link.
The page looks almost identical to the company's login page.
He enters with username and password.
Rahul does not notice that the website is controlled by the attacker.
The attacker now has Rahul's login credentials.

Stage I- Unauthorised Access
The attacker uses Rahul's credentials to access BrightMart's internal system.
Because Rahul uses the same password for another system, the attacker is also able to access that system.
The attacker discovers that Rahul has access to Customer information.
The attacker downloads a large amount of customer data.
what cybersecurity problem has occured?

Stage II - Malware Attack

The attacker sends another e-mail to Rahul containing a malicious attachment.

Rahul opens the attachment.

Malware is installed in his computer.

The malware begins communicating with the attacker's system.

The attacker now attempts to move from Rahul's computer to other systems within BrightMart network.

Which vulnerability allowed the attacker to move further into his organisation?

Stage III - Ransomware

Two days later, employees arrive at work and discover that they cannot access their important files.

A message appears on their computers:

"Your files have been encrypted. Pay the Ransom to recover them. The company's customers database and several important business files are encrypted."

Unfortunately, the company's most recent backup is almost a month old.

BrightMart website stopped working.

Customer cannot place orders.

The customer support team cannot access customer records.

Stage IV - Business Impact
BrightMart now faces several problems.
Financial Impact:-
The company loses sales because its website is unavailable.
Data Impact:-
customer Information may have been stolen.
Operational Impact:-
Employees cannot access important systems.

Reputation Impact:-
Customers loose confident in the company.
legal/compliance Impact:-
The company may face regulatory or legal consequences depending on what information was exposed according to the laws.

Group 1 - Identify the Threats.
what threats are present in this case?
Group 2 - Identify the vulnerabilities.
Which weakness in BrightMart's security allowed the attack to succeed?
Group 3- Apply the CIA Triad.
Was confidential information exposed?
Was information changed or damaged?
Were the systems available to legitimate users?
Group 4 - Security Solutions.
who is responsible for cybersecurity?
was phishing the only problem?
What was the biggest vulnerability?

Solution:-

Case Study-1

Q. What cybersecurity problem has occured?

Ans:
- Phishing
- Credential Theft
- Unauthorised access
- Data Breach

Q. Which vulnerablity allowed the attacker to move further into the organisation?

Ans:
- Weak Passwords
- Excessive user privileges
- Lack of Multi-Factor-Authentication.
- Poor network security
- Lack of employee awareness
- Outdated Software

Q. What threats were present in this case?

Ans:
- Phishing
- Malware
- Ransomware
- Credential Theft
- Unauthorised Access
- Data Theft.

Q. What weaknesses in BrightMart's security allowed the attack to succeed?

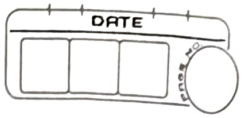
Ans:

Vulnerability	How it helped the attacker
Weak Passwords	• Easier to comprise accounts
Password Reuse	One stolen password gives access to multiple systems.
No MFA	Stolen Password may be enough to login.
Lack of Training	Employee falls for phishing.
Delayed Updates	Vulnerable software may be exploited.
Infrequent Backups	Difficult to recover data after ransomware

→ Confidentiality
Was confidential information exposed?
Yes,
Customer information was accessed by the attacker.

→ Integrity
Was information changed or damaged?
Potentially Yes,
The attacker could modify or encrypt the files.

→ Availability
Was systems available to legitimate users?
No,
The website and internal systems became unavailable due to ransomware.



So,
Confidentiality → Affected
Integrity → Affected
Availability → Affected.

Group 4 - Security Solutions
What should BrightMart do to prevent this from happening again?

- Use strong passwords
- Implement Multi factor Authentication (MFA)
- Provide regular cybersecurity awareness Training
- Keep software updated
- Follow the principle of least privilege
- Segment important networks
- Encrypt sensitive data
- Regular Backups
- Develop an incident response plan

Q. Who is responsible for cybersecurity?
Ans. Not only the IT department. Every employee in the organisation, systems has a role in it.

Q. Was phishing the only problem?
Ans. No,
Phishing was the initial attack, but weaknesses such as password reuse, lack of MFA, excessive privileges & poor backups allowed the attack to become much more damaging.

Q. What was the biggest vulnerablity?

Ans. "lack of employee awareness was a major vulnerablity because the attack started when Rahul clicked the phishing link."

Quick Assesment

Q. What was the initial attack against BrightMart?

Ans. Phishing,
Rahul recieved a fake IT email and entered his username and password on a fraudent login page.

Q. Give 3 vulnerablities that helped the attacker.

Ans.
- Employees used simple passwords
- Employees reused the same passwords
- MFA was not enabled for some important systems.

Q. Which 3 security Principles of CIA Triad were were affected?

Ans.
- Confidentiality
- Integrity
- Availability

Q. What was the impact of Ransomware on BrightMart?

Ans. Ransomware encrypted important files, stopped the website and prevented employees from accessing internal essential systems.

Q. Suggest 3 security controls Bright Mart should implement:

Ans
- MFA for important accounts.
- Regular cybersecurity training for employees.
- Regular Backups.

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